

Total no. of Pages: 2

Roll no.

Degree: B-Tech.

Semester 3rd

END-SEMESTER EXAMINATION, June, 2026

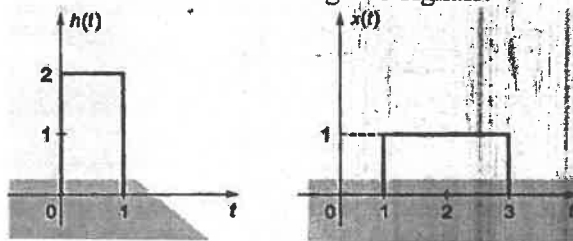
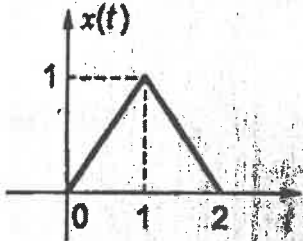
Course Title: Signals and Systems

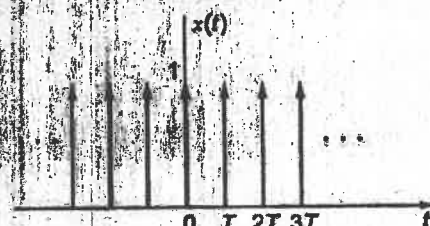
Course Code: Sem: ECECC302/BIECC302/VTECC302

Duration: 3.00 Hours

Max. Marks: 50

Note: - Attempt all questions in the given order only. Missing data/information (if any), may be suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
1	<i>Attempt any two questions of the following</i>		
1a	Determine whether or not each of the following systems are time variant with input $x(t)$ and output $y(t)$. (i) $y(t) = tx(t)$ (ii) $y(t) = tx(t) + 4$ (iii) $y(t) = x(2t)$	5	C01
1b	Discuss step response of an LTI system in detail. If step response is $y(t) = 2e^{-t}u(t)$, calculate its impulse response.	5	C01
1c	Determine whether or not each of the following systems are invertible with input $x(t)$ and output $y(t)$. (i) $y(t) = x^2(t)$ (ii) $y(t) = x(2t)$ (iii) $y(t) = \frac{d}{dt} x(t)$	5	C01
2	<i>Attempt any two questions of the following</i>		
2a	Find the convolution of the following two signals. 	5	C02
2b	Draw the even and odd components of the waveform of a signal $x(t)$ shown in figure. 	5	C02
2c	Discuss the terms stability and causality of a system using mathematical expressions.	5	C02

3	Attempt any two questions of the following		
3a	Find the Fourier series of the following signal. 	5	C03
3b	State and prove Parseval's theorem	5	C03
3c	Find the Fourier transform of the following signals: (i) $x(t) = \frac{1}{1+t^2}$ (ii) $x(t) = \frac{1}{1+jt}$ (iii) $x(t) = \frac{1}{\pi t}$	5	C03
4	Attempt any two questions of the following		
4a	Find the Laplace transform of the following signals and corresponding ROC. (i) $x(t) = e^{-t}u(t) + e^{-2t}u(t)$ (ii) $x(t) = e^{-t}u(t) + e^{2t}u(-t)$ (iii) $x(t) = e^t u(t) + e^{-2t} u(-t)$	5	C04
4b	Given that $X(s) = \frac{s^2+12}{s(s+2)(s+3)}$. Find the inverse Laplace of X(s).	5	C04
4c	The impulse function of the system is given by $H(s) = \frac{2}{s+1} + \frac{1}{s-1}$. Determine the impulse response if the system is (i) Causal (ii) Stable	5	C04
5	Attempt any two questions of the following		
5a	Given that $x[n] = a^n u[n]$. Find the Z-transform of the following signals (i) $g[n] = nx[n]$ (ii) $g[n] = [n+1]x[n+1]$ (iii) $g[n] = x[n]*x[n]$	5	C05
5b	Discuss the relationship between Z-transform and Laplace transform	5	C05
5c	Solve the difference equation $y[n] + 3y[n-1] = x[n]$ with initial conditions $y[-1]=1$ and determine $y[n]$ for the input $x[n] = 8 u[n]$.	5	C05